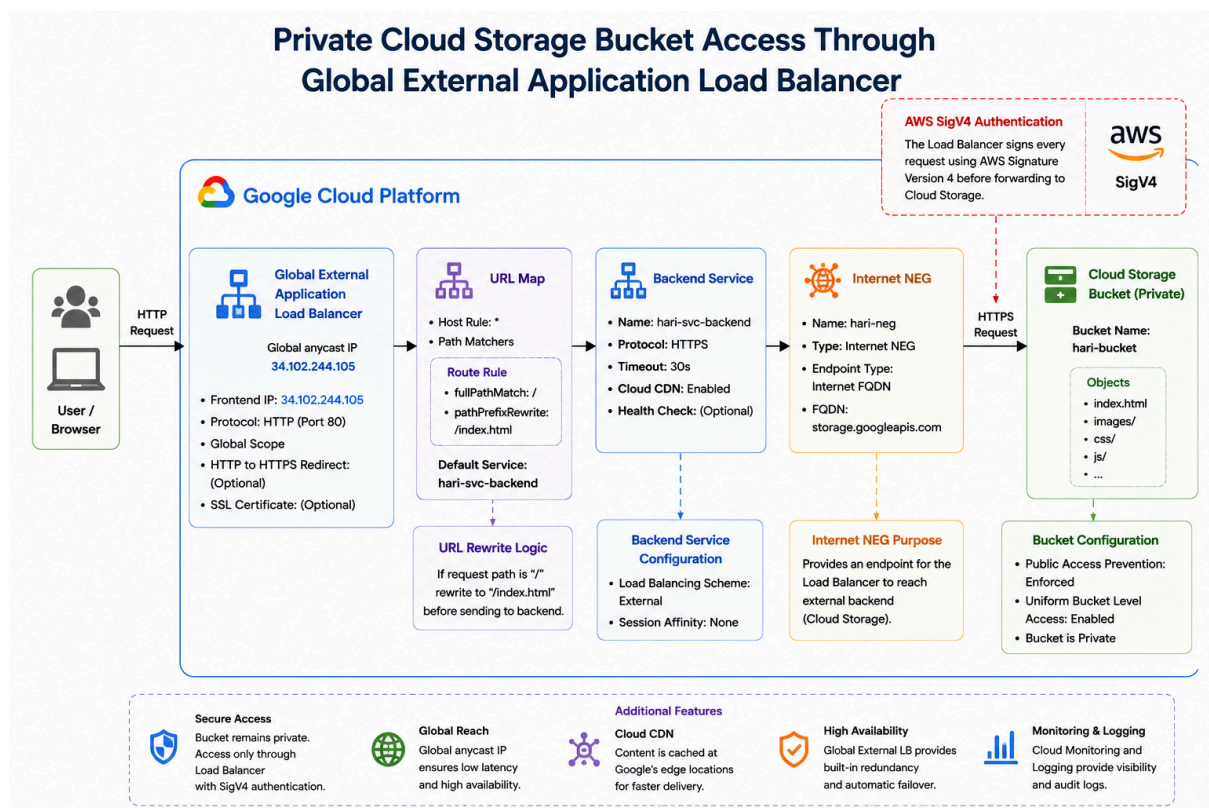


Private Cloud Storage Bucket Access Through Global External Application Load Balancer

Architecture Diagram



Step 1: Create Cloud Storage Bucket

Description

Created a Cloud Storage bucket to host static website content.

Action Performed

- Bucket Name: hari-testing
- Region: us-central1
- Storage Class: Standard

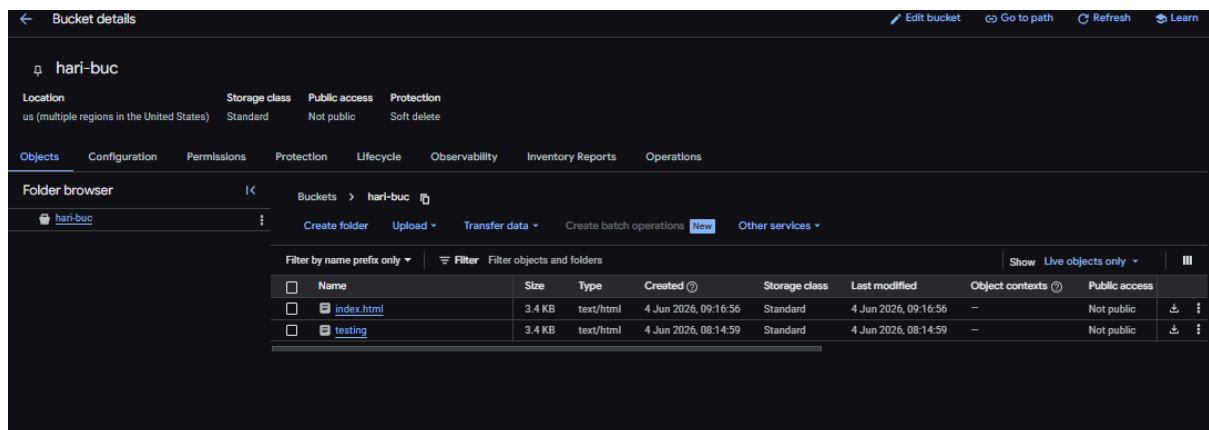
Purpose

To store website files.

Function

Acts as the backend content repository.

Screenshot



Validation

Bucket created successfully.

Step 2: Upload Website Content

Description

Uploaded the website files into the bucket.

Action Performed

Shell

```
gcloud storage cp index.html gs://hari-testing
```

Purpose

To provide content for the load balancer to serve.

Function

Stores static website assets.

Screenshot

Screenshot: Bucket Objects List

Validation

index.html visible in bucket.

Step 3: Make Bucket Private

Description

Configured bucket to prevent public access.

Action Performed

- Uniform Bucket Level Access Enabled
- Public Access Prevention Enabled

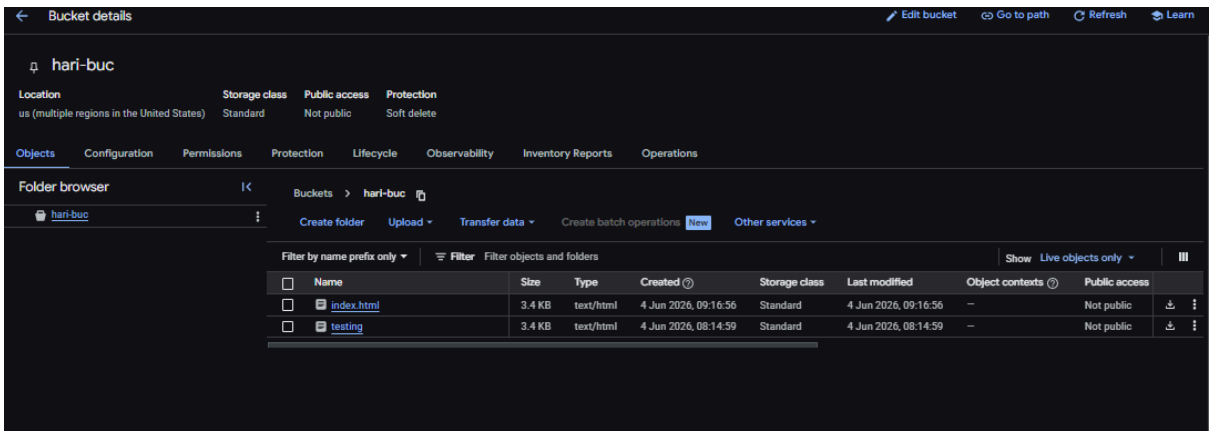
Purpose

To secure objects from direct internet access.

Function

Only authenticated services can access objects.

Screenshot



Validation

Anonymous access denied.

Step 4: Create Internet NEG

Description

Created Internet Network Endpoint Group.

Action Performed

- NEG Name: hari-neg

- Endpoint Type: Internet

Purpose

To connect Cloud Load Balancer with Cloud Storage.

Function

Provides external backend endpoint.

Screenshot

The screenshot shows the 'Network endpoint group details' page for a resource named 'hari-neg'. The page has a dark theme. At the top, there's a navigation bar with a back arrow, the title 'Network endpoint group details', and a 'Delete' button. Below this, the resource name 'hari-neg' is displayed. A table lists the following details:

Network endpoints	1
Network endpoint group type	Internet network endpoint group (external backends)
Scope	Global
Default port	443
In use by	hari-svc-backend
Creation time	Jun 4, 2026, 8:21:06 am UTC+05:30
Tags	-

Below the table, there's a section titled 'Network endpoints in this group'. It contains a descriptive paragraph: 'Network endpoints represent your services (applications, load balancing) and diverse infrastructure (VM instances, containers, etc) in a standard manner regardless of their location. [Learn more](#) '. Underneath this text are two buttons: 'Add network endpoint' and 'Remove endpoint'.

At the bottom, there's a 'Filter' section with the text 'Filter by IP or port'. To the right of the filter text are a help icon and a menu icon. Below the filter is a table with the following structure:

☐	IP address/Fully qualified domain name	Port	
☐	hari-buc.storage.googleapis.com	443	

At the very bottom of the screenshot, there is a link that says 'Equivalent REST'.

Validation

NEG created successfully.

Step 5: Configure AWS SigV4 Authentication

Description

Configured backend authentication.

Action Performed

- AWS Signature Version 4 Enabled
- Access Key Configured
- Secret Key Configured

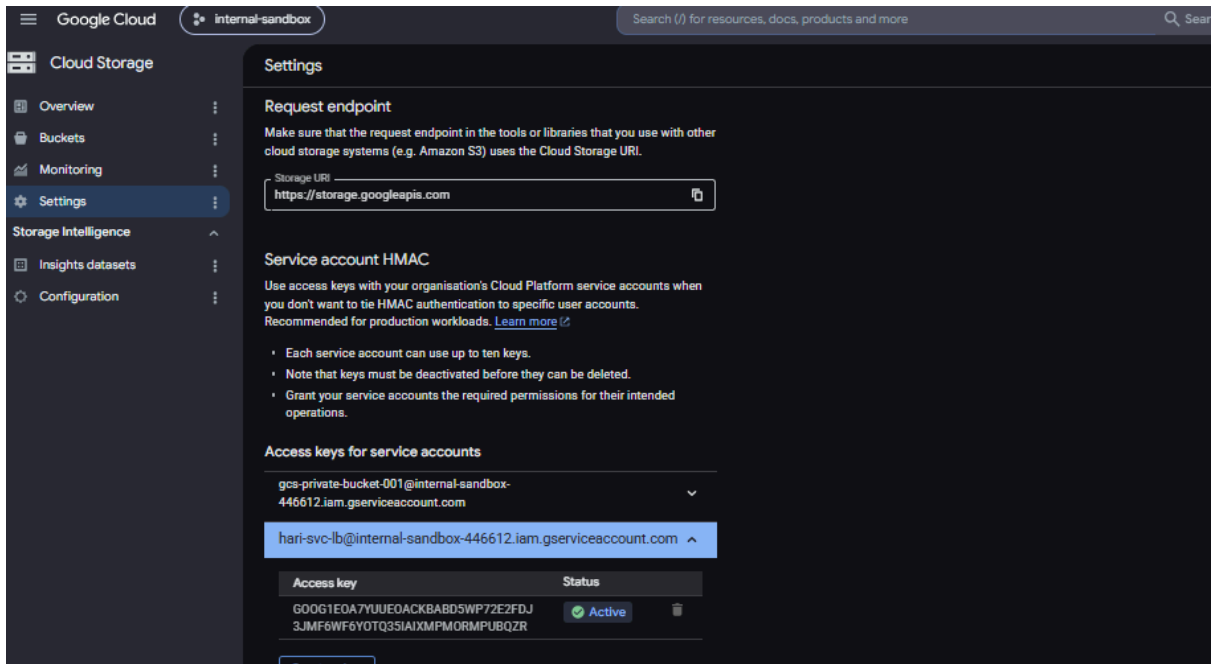
Purpose

To authenticate requests sent to the private bucket.

Function

Signs requests before reaching Cloud Storage.

Screenshot



Validation

Backend authenticated successfully.

Step 6: Create Backend Service

Description

Created backend service and attached Internet NEG.

Action Performed

- Backend Service: hari-svc-backend
- Protocol: HTTPS
- Backend: hari-neg
- Cloud CDN: Enabled

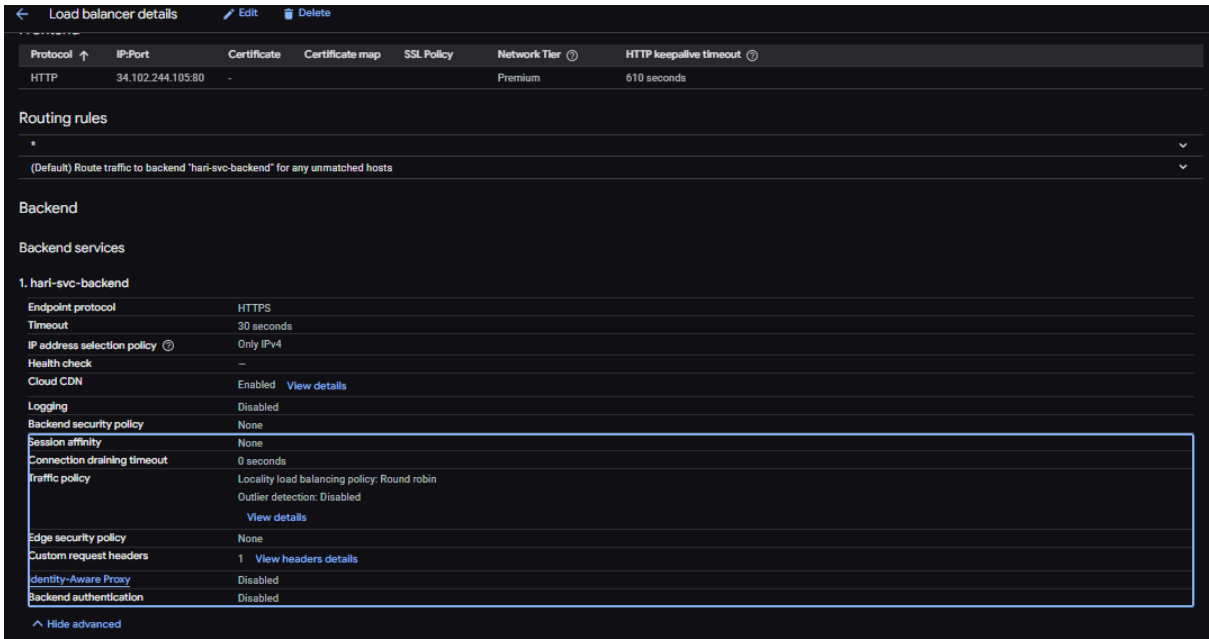
Purpose

To route traffic toward Cloud Storage.

Function

Acts as middle layer between LB and bucket.

Screenshot



Validation

Backend service healthy.

Step 7: Create URL Map

Description

Created URL Map for request routing.

Action Performed

- URL Map Name: hari-svc-lb
- Default Backend: hari-svc-backend

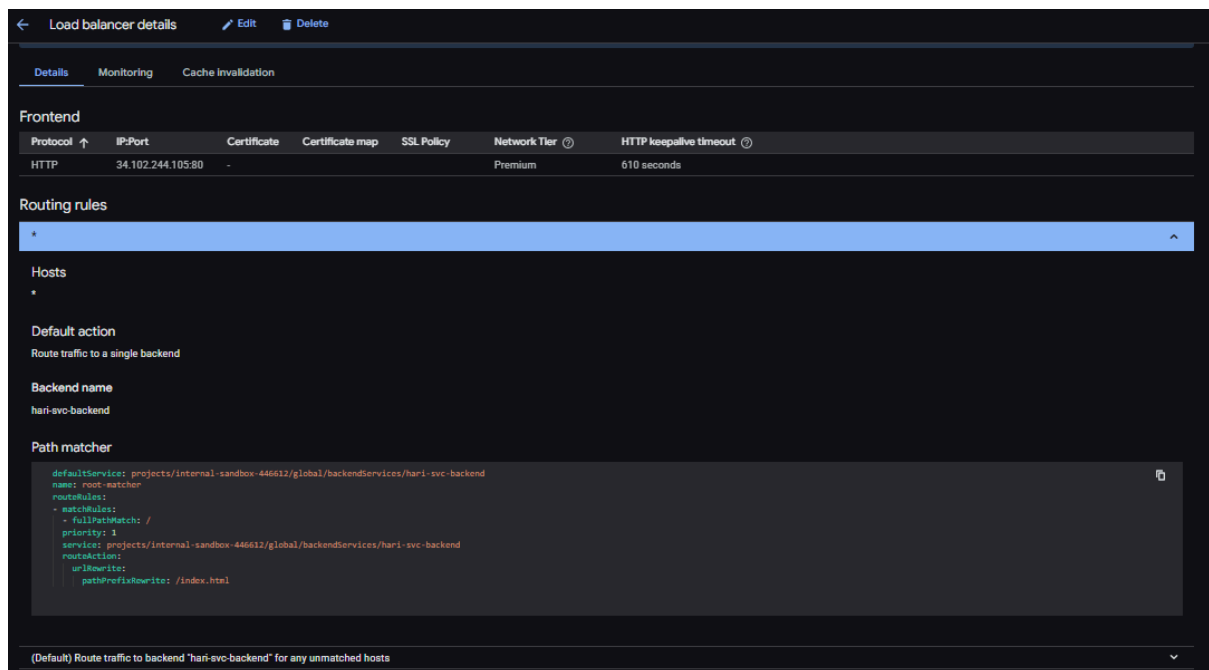
Purpose

To control request routing.

Function

Maps incoming requests to backend service.

Screenshot



Validation

URL Map created successfully.

Step 8: Create Target HTTP Proxy

Description

Created HTTP Proxy.

Action Performed

- Proxy Name: hari-http-proxy

Purpose

To connect URL Map with Forwarding Rule.

Function

Processes incoming HTTP requests.

Screenshot

Screenshot: HTTP Proxy Configuration

Validation

Proxy active.

Step 9: Create Forwarding Rule

Description

Created frontend listener.

Action Performed

- Protocol: HTTP
- IP Address: 34.102.244.105

Purpose

To expose service publicly.

Function

Receives client requests.

Screenshot

Screenshot: Forwarding Rule

Validation

External IP assigned successfully.

Step 10: Initial Testing

Description

Tested load balancer using external IP.

Action Performed

Shell

```
curl http://34.102.244.105/
```

Result

Received error.

Screenshot

Screenshot: Browser Error Page

Issue 1: NoSuchKey Error

Error Observed

XML

```
<Error>  
<Code>NoSuchKey</Code>  
</Error>
```

Root Cause

The load balancer forwarded:

None

/

to Cloud Storage.

Cloud Storage searched for:

None

bucket/

However only:

None

index.html

existed.

No object named "/" exists.

Impact

Website could not load.

Resolution

Implemented URL Rewrite Rule.

Step 11: Create URL Rewrite Configuration

Description

Configured root request rewriting.

Action Performed

Created:

None

defaultService:

<https://www.googleapis.com/compute/v1/projects/internal-sandbox-446612/global/backendServices/hari-svc-backend>

```
hostRules:
- hosts:
  - "*"
  pathMatcher: root-matcher

pathMatchers:
- name: root-matcher
  defaultService:
https://www.googleapis.com/compute/v1/projects/internal-sandbox-446612/global/backendServices/hari-svc-backend

routeRules:
- priority: 1

matchRules:
- fullPathMatch: /

routeAction:
  urlRewrite:
    pathPrefixRewrite: /index.html

service:
https://www.googleapis.com/compute/v1/projects/internal-sandbox-446612/global/backendServices/hari-svc-backend
```

Purpose

Serve index.html when users request root URL.

Function

Rewrites:

None

/

to

None

/index.html

before forwarding request.

Screenshot

Screenshot: root-rewrite.yaml

Issue 2: URL Map Validation Errors

Error Observed

Several YAML syntax and URL Map import failures occurred.

Examples:

None

Invalid field pathMatchers

RouteRule not properly defined

Import failed

Root Cause

Incorrect YAML structure and nesting.

Resolution

Validated configuration using:

```
Shell
gcloud compute url-maps validate \
--source=root-rewrite.yaml
```

until:

```
None
loadSucceeded: true
testPassed: true
```

was returned.

Screenshot

Screenshot: Validation Success

Step 12: Import Updated URL Map

Description

Applied updated routing rules.

Action Performed

```
Shell
gcloud compute url-maps import hari-svc-lb \
--source=root-rewrite.yaml \
--global
```

Screenshot

Screenshot: Import Success

Validation

Import completed successfully.

Issue 3: Old Content Served From CDN Cache

Error Observed

Website still showed old behavior after URL Map update.

Root Cause

Cloud CDN cached previous responses.

Resolution

Invalidated CDN cache.

Shell

```
gcloud compute url-maps invalidate-cdn-cache \  
hari-svc-lb \  
--path="/*"
```

Screenshot

Screenshot: CDN Cache Invalidation

Validation

Fresh content retrieved from backend.

Step 13: Final Testing

Description

Verified website accessibility.

Action Performed

Shell

```
curl -v http://34.102.244.105/
```

Result

HTML

Backend Services

page loaded successfully.

Screenshot

Screenshot: Successful Browser Output

Validation

HTTP 200 OK returned.

Step 14: Browser Validation

Description

Validated access through browser.

Action Performed

Opened:

None

http://34.102.244.105

Result

Backend Services page displayed.

Screenshot

Screenshot: Final Working Page

(Use the screenshot you shared where the Backend Services table is visible.)

Lessons Learned

Issue	Root Cause	Solution
NoSuchKey	Root path "/" not mapped to object	URL Rewrite
URL Map Import Failure	Invalid YAML structure	Validate and re-import
Old Content Displayed	Cloud CDN cache	Cache Invalidation

Bucket Access Issues	Private bucket authentication required	AWS SigV4
Direct Bucket Access Failed	Bucket intentionally private	Access through Load Balancer

Project Outcome

Successfully Implemented

- Private Cloud Storage Bucket
- Internet NEG
- Backend Service
- Global External Application Load Balancer
- AWS SigV4 Authentication
- Cloud CDN
- URL Rewrite Rule
- Root URL → index.html Mapping
- HTTP 200 Response
- Browser Access Verified

Final Result

Users can access:

None

<http://34.102.244.105/>

while the Cloud Storage bucket remains private and protected. This demonstrates secure content delivery using a Global External Application Load Balancer with Internet NEG and URL Rewrite functionality.